

Infosys SpringBoard Virtual Internship 6.0

Internship Completion Report Batch 11

Group 1

Team Name : Team B

Team Members : Arya Mishra , Kritika Balpande , Radha sri Yakkala

Mentor Name : Mrs. Nithyasri S J

Internship Duration : 8 weeks

Year : 2026

1. Project Title :

“ElectViz - Election data Visualization for Media”

2. Project Objective:

- To analyze election data across years, states, and constituencies.
- To study party performance, vote share, and seat dominance.
- To understand voter turnout and gender representation.
- To analyze electoral competitiveness using victory margin and winner strength.
- To present insights through interactive and user-friendly dashboards.

3. Project Description:

3.1 Project Overview

India has the world's largest democracy, where elections are conducted across many states with thousands of candidates and millions of voters. Because of this scale, it is important to analyze election data from multiple perspectives

Electviz is a data visualization project developed using power BI .

Electviz transforms large scale election data into interactive and informative dashboards that present clear analysis and insights from raw election data in an engaging manner , thereby improving clarity, transparency, and understanding.

3.2 Project Approach:

1. Problem Identification:

- The analysis and interpretation of election data pose significant challenges due to the large volume, complexity, and multidimensional structure of electoral datasets.
- Election data consists of numerous attributes such as constituencies, candidates, political parties, vote counts, victory margins, and regional classifications. When this data is presented in raw or tabular formats, it becomes difficult to derive meaningful insights, limiting its usefulness for accurate and data-driven political reporting.
- Media organizations and analysts often require timely, clear, and visually intuitive representations of such data to communicate election outcomes effectively to the public.
- The problem addressed by the ElectViz project is the absence of an interactive and visual analytical system that can simplify complex election datasets while maintaining accuracy and depth of analysis.
- Using Power BI, the project aims to transform raw election data into dynamic dashboards that highlight key indicators such as voting trends, party performance, and regional election dynamics. The solution involves data cleaning, preprocessing, and the identification of relevant metrics to ensure reliable analysis.
- By leveraging Power BI's visualization and interactivity features, the project enables users to explore election data efficiently, supporting transparent, fact-based interpretation and enhanced political reporting.

2. Data Collection:

The dataset used for this project was sourced from Kaggle, titled “Indian Election Dataset” by Nanadan Pandey.

Link <https://www.kaggle.com/datasets/awadhi123/indian-election-dataset>

This dataset contains detailed candidate-level election data for India's national and state legislative elections, specifically the Lok Sabha elections. It covers election years from 1977 to 2015, with each record representing an individual candidate contesting in a specific constituency during a given election year.

The Dataset contains 11 columns and 73080 rows.

Column Name	Description
st_name	Name of the state
Year	General election year
pc_no	Parliamentary constituency number
pc_name	Name of the parliamentary constituency
pc_type	Reservation status of the constituency
cand_name	Name of the contesting candidate
cand_sex	Gender of the candidate
partyname	Name of the political party
partyabbre	Abbreviation of the political party
totvotpoll	Total votes received by the candidate
electors	Total number of registered voters in constituency

3. Data Cleaning and Preprocessing:

The first step is to clean the dataset and perform early preprocessing so the data is uniform and dashboard made and correct and insightful. Steps performed are:

1. Checking the data types of each column; all data types were correctly detected by PowerBI data loader
2. Checking for null /empty values ; Column pctype there are 8070 null values and candidate gender also contains 542 null values . So null values were replaced with unknown .36 blank values of state were replaced by Goa.
3. Checking for duplicates; selected all columns and entries and then used remove duplicate rows feature in the power query editor
4. Replaced mismatching or different spelling values in 'state name' column .
 - Chattisgarh corrected to Chhattisgarh
 - Goa , Daman & Diu replaced to Goa
 - Goa Daman & Diu replaced to Goa
 - Odisha replaced to Orissa
 - Uttaranchal replaced to Uttarakhand

- Pondicherry replaced to pundicherry
 - Nct Of Delhi replaced to Delhi
 - National Capital of territory to Delhi
5. Replaced mismatching or different spelling values in 'party_name' column like BJP and Bhartiya Janta Party and similarly 9 values. Got 45 distinct values reduced to 36 distincts.

For example

- Bharatiya Janta Party , BJP replaced to Bharatiya Janata Party
 - Independents replaced to Independent
 - Janata Pary replace to Janata Party
 - Janta Party(JP) replaced to Janta Party
 - BSP replaced to Bhahujan Samaj Party
 - INC replaced to Indian National Congress
 - SP replaced to Samajwadi Party
 - Akhil Bharatiay Ramrajya Parishal replaced to Akhil Bharatiya Ramrajya parishad
 - AAP replaced to Aam Aadmi Party
 - Republican Party of India (Khobragade) corrected to Replublican Party Of India
 - AASP replaced to Ashankya Samaj Party
 - AIFB replaced to All India Forward Bloc
 - AITC replaced to All India Trinamool Congress
 - NCP replaced to Nationalist Congress of India
 - IND replaced to Independent
 - CPM replaced to Communist Party of India(Marxist)
6. Added a custom column 'voters_turnout_%' to calculate percentage of votes received over total voters and winner flag 1 for winners and 0 for non winners.
- $\text{turnout\%} = (\text{votes received} / \text{Total voters registered}) * 100$
7. Changed the type of turnout column to percentage type.

4. Data Modelling

- Created a Winner Summary table from the main dataset by grouping data based on Parliamentary Constituency Number and Year, and calculating the maximum votes received in each group.
- Merged the Winner Summary table back with the main table using three matching columns:Parliamentary Constituency Number , Year and Votes Received.
- Expanded only the required column from the merged table.
- Created a custom Winner column using Conditional Column:
 - If merged value is *null* → Winner = 0
 - If merged value contains maximum votes → Winner = 1

5. Measure Creation using DAX

- Total Candidates
- Total Female Candidates
- Total Male Candidates
- Total States
- Total Parliamentary Constituencies
- Total Parties
- Total Constituency Votes
- Total Registered Voters
- Actual Votes
- Seats Won
- Total Parliamentary Constituency number
- Victory Margin %
- Winner Strength %

6. Dashboard Design and Insight Generation

Created a simple and elegant dashboard using different visuals to understand election data from multiple perspectives.

1. Page 1 Election Overview Page
2. Page 2 Year wise Analysis
3. Page 3 State wise analysis
4. Page 4 Distribution of parliamentary Constituency
5. Page 5 Party performance analysis
6. Page 6 Gender Demonstration
7. Page 7 Candidate Performance Analysis
8. Page 8 Electoral competitiveness and winner analysis

3.3 Technology Used:

- Power BI Desktop - used for dashboard creation and data visualization
- Power Query Editor - Used for data cleaning, transformation, handling null values, grouping, and merging operations.
- DAX (Data Analysis Expression) - Used for creating calculated measures, KPIs, victory margin %, winner strength %, and dominance analysis.
- Microsoft word : Used for writing documentation.
- Microsoft Teams - Used for project discussions, meetings, and coordination.

- Microsoft Excel - for storing dataset
- Github and Gitlab - Used for version control, code management, and collaboration.

3.4 Key Insight

- There is gender imbalance in Indian elections i.e. male dominance although Female participation increases over time but still low .
- The majority of parliamentary constituencies belong to General , followed by ST and SC but ST and SC play an important role in voter participation by showing higher voter turnout%.
- There was political instability in the year 1996-1998 .In the 1996-1998 count of the total number of parties and total number of candidates decreased and then started increasing from 1999. In 1996 , many regional parties which contested independently formed coalition with other parties in 1998 this can be one of the reasons for the decline.
- Winner Strength % and Victory margin % across the year are fluctuating which tells that in some years elections are highly dominating or in some years elections are less dominating.
- National parties like INC and BJP dominate vote share, but regional parties still play a significant role.
- Some parties win more seats even with small vote shares, and some parties win low seats with high vote share showing high seats does not mean high votes received.

3.5 Real World Impact and Public Communication

- Dashboards break down the complex raw data into visuals that are easier for the general public to understand.
- Helps parties to make the strategy . Strategists use these insights to decide not just where a candidate should hold a rally , and what specific issues focus on.
- Dashboards allow anchors to instantly compare current turnout% or margin with past elections.
- The “what-if” scenario : Anchors use interactive sliders to show “If the BJP loses 20 seats from UP , they must gain 10 from West Bengal and 10 in Odisha to stay on track.

4. Timeline Overview

Week	Activities Planned	Activities Executed
Week 1	Dataset collection and understanding	Imported election dataset containing variables like candidate names, party names, gender, and vote counts.
Week 2	Data cleaning and transformation	Verified data types for votes received and Winner columns to ensure they are ready for aggregation.
Week 3	Data modeling and DAX measure creation	Created basic count measures for seats and candidates; organized fields into a measureTable for better management.
Week 4	Dashboard development Page 1 Election overview Page 2 Party Performance Analysis Page 3 Distribution of parliamentary constituency	Created insight full dashboard using different visuals to show election overview, to understand party performance and distribution of parliamentary constituency type.
Week 5	Dashboard development Page 4 Year wise analysis Page 5 State Wise Analysis	Dashboard creation using different visuals to year and state wise analysis.
Week 6	Dashboard development Page 6 Gender Wise Analysis Page 7 Candidate Performance Analysis Page 8 Electoral Competitiveness and winner	Created a simple and elegant dashboard using different visuals to understand gender representation, candidate performance and

	Analysis	competitiveness .
Week 7	Insight validation and documentation	Finalized visual formatting (borders/shadows) and generated a comprehensive project document detailing the logic used.
Week 8	Demonstration of working project	A live demonstration showing the completed dashboard and explaining the insight derived from each page.

5. Milestone

5.1 Key Milestones:

MILESTONE NAME	DATE COMPLETED	WORK DONE	INSIGHT GENERATED
MILESTONE 1	16/01/2026	Collected Indian parliamentary election dataset and Preprocessed the data .	Conversion of raw data into analysis ready format.

MILESTONE 2	23/01/2026	Dashboard creation started , three dashboard are created 1. Election overview 2. party performance analysis 3. Distribution of parliamentary constituency type.	• Voter participation and total votes have increased over the years, showing growing political engagement, but votes remain concentrated among a few dominant national parties like INC and BJP. • General constituencies dominate in number , SC and ST constituencies play a crucial role in representation and demonstrate voter participation. • Gender imbalance remains significant, as male candidates dominate across all constituency types, while female participation is increasing slowly.
MILESTONE 3	30/01/2026	Two more Dashboards are created 1. Year wise analysis 2. State wise analysis.	• There was the highest turnout % in 1977 then it kept on decreasing and lowest in 1996 then it increased from 1998 and then became stable in 2014. It tells changes in public engagement over time. • There is male dominance in winning candidates compare to female winning candidates. • Increases in total votes do not consistently align with higher turnout percentages, indicating no clear positive

			relationship between voter participation and total votes.
MILESTONE 4	06/02/2026	Three Dashboard are created 1. Gender wise analysis 2. candidate performance analysis 3. Electoral competitiveness.	• Gender imbalance continues in Indian elections, with male candidates dominating both in numbers and votes, while female participation shows only gradual improvement. • Votes are concentrated among a few strong candidates and parties, indicating uneven competition and consistent voter trust in certain leaders. • Electoral competitiveness varies across years, as victory margins and winner strength fluctuate, and higher seat share does not always mean higher vote share under the First-Past-The-Post system.

5.2 Project Execution Details

The project was executed in four milestones, starting from data preparation and ending with interactive electoral analysis dashboards.

- Milestone 1: Collected Indian parliamentary election dataset and Preprocessed the data . Conversion of raw data into analysis ready format.

- Milestone 2 :Dashboard creation started , three dashboards are created i.e., election overview , party performance analysis and distribution of parliamentary constituency type.
- Milestone 3:Two more Dashboards are created on year wise analysis and state wise analysis.
- Milestone 4:Three Dashboards are created on gender wise analysis ,candidate performance analysis, and electoral competitiveness.

1. Problem Identification:

- a. The analysis and interpretation of election data pose significant challenges due to the large volume, complexity, and multidimensional structure of electoral datasets.
- b. The problem addressed by the ElectViz project is the absence of an interactive and visual analytical system that can simplify complex election datasets while maintaining accuracy and depth of analysis.

2. Data Collection:

- a. The dataset used for this project was sourced from Kaggle, titled “Indian Election Dataset” by Nanadan Pandey. Link <https://www.kaggle.com/datasets/awadhi123/indian-election-dataset>
- b. This dataset contains detailed candidate-level election data for India’s national and state legislative elections, specifically the Lok Sabha elections.

3. Data Preprocessing

- a. The first step is to clean the dataset and perform early preprocessing so the data is uniform and dashboard made and correct and insightful. Steps performed are:
- b. Checking the data types of each column; all data types were correctly detected by the PowerBI data loader.
- c. Checking for null /empty values ; Column pctype there are 8070 null values and candidate gender also contains 542 null values . So null values were replaced with unknown .36 blank values of state were replaced by Goa.
- d. Checking for duplicates; selected all columns and entries and then used remove duplicate rows feature in the power query editor.
- e. Replaced mismatching or different spelling values in ‘state name’ column .
- f. Replaced mismatching or different spelling values in ‘party_name’ column like BJP and Bhartiya Janta Party and similarly 9 values. Got 45 distinct values reduced to 36 distincts.
- g. Added a custom column ‘voters_turnout_%’ to calculate percentage of votes received over total voters and winner flag 1 for winners and 0 for non winners.
- h. Changed the type of turnout column to percentage type.

4. Data Modelling

- Created a Winner Summary table from the main dataset by grouping data based on Parliamentary Constituency Number and Year, and calculating the maximum votes received in each group.
- Merged the Winner Summary table back with the main table using three matching columns: Parliamentary Constituency Number , Year and Votes Received.
- Expanded only the required column from the merged table.
- Created a custom Winner column using Conditional Column:
 - If merged value is *null* → Winner = 0
 - If merged value contains maximum votes → Winner = 1

5. Dashboard Creation and DAX measures:

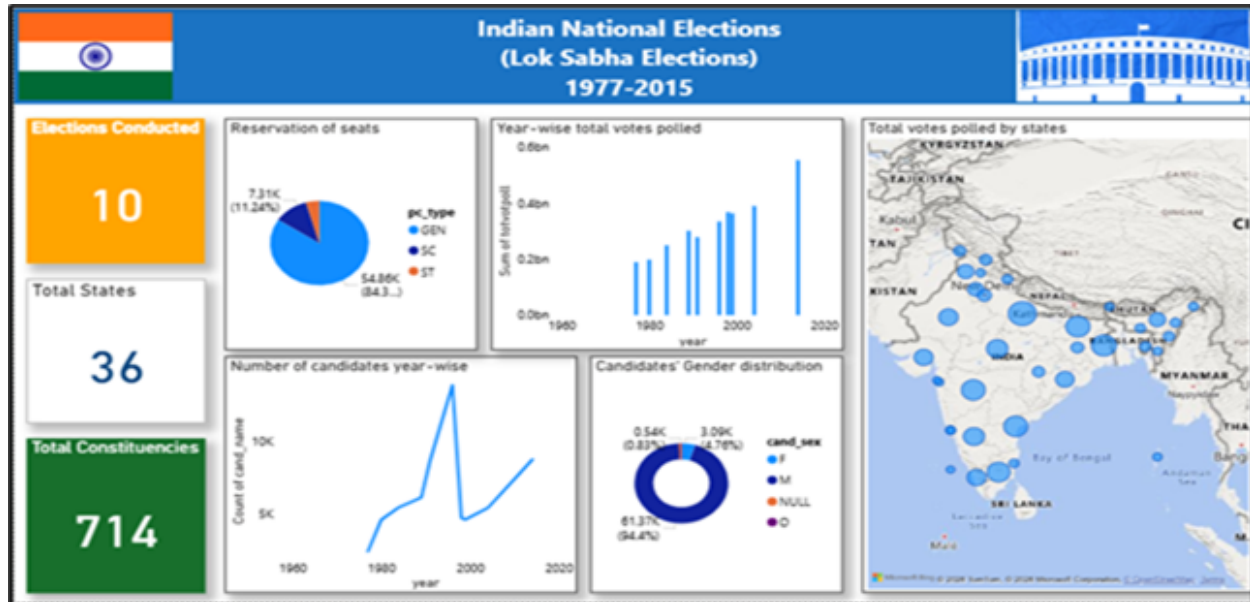
- a. Created a simple and elegant dashboard using different visuals to understand election data from multiple perspectives.
 - i. Page 1 Election Overview Page
 - ii. Page 2 Year wise Analysis
 - iii. Page 3 State wise analysis
 - iv. Page 4 Distribution of parliamentary Constituency
 - v. Page 5 Party performance analysis
 - vi. Page 6 Gender Demonstration
 - vii. Page 7 Candidate Performance Analysis
 - viii. Page 8 Electoral competitiveness and winner analysis

6. Documentation and presentation

- a. Presentation of a working dashboard and explaining the insights derived from different pages.

6.Snapshots/Screenshot

6.1 Election Overview (1977–2015)



Page Objective

This page provides a **high-level overview of Indian Lok Sabha elections**, helping us understand the **scale, participation, and growth of elections over time** before going into deeper analysis.

Data used : Year , state , votes received , turnout % , candidate gender , parliamentary constituency type

Visuals used: KPI Cards (to get quick summary of total states and parliamentary constituency) , Line Chart (trend) ,bar graph (comparison) , Pie Chart and Donut Chart (Distribution) , Map (to get votes received by each state)

1. Card: Count of State Name (35)

What it shows:

This card shows the total number of states involved in elections.

Insight:

Elections are conducted across a wide geographic area, nationwide nature of Indian democracy.

2. Card: Count of Parliamentary Constituencies (717)**What it shows:**

Displays the total number of parliamentary constituencies covered in the dataset.

3. Card: Count of Years (11)**What it shows:**

Shows the total number of election years analyzed (1977–2014).

Insight:

This allows us to study long-term trends and changes in voting behavior and participation.

4. Pie Chart: Distribution of Parliamentary Constituency Type**What it shows:**

Breakdown of constituencies into GEN, SC, and ST categories.

Insight:

Most constituencies are General, while SC and ST seats ensure constitutional representation for marginalized communities.

5. Column Chart: Total Votes Received by Year**What it shows:**

Total votes polled in each election year.

Insight:

An increasing trend indicates rising voter participation and population growth over time.

6. Map: Sum of Votes Received by State

What it shows:

Geographical distribution of votes across Indian states.

Insight:

Larger states contribute significantly more votes, highlighting regional voting weight in national outcomes.

7. Line Chart: Count of Candidates per Year

What it shows:

Number of candidates contesting elections in each year.

Insight:

A rise in candidates reflects increased political competition and democratic participation.

8. Donut Chart: Gender Distribution of Candidates

What it shows:

Proportion of male, female, and other candidates.

Insight:

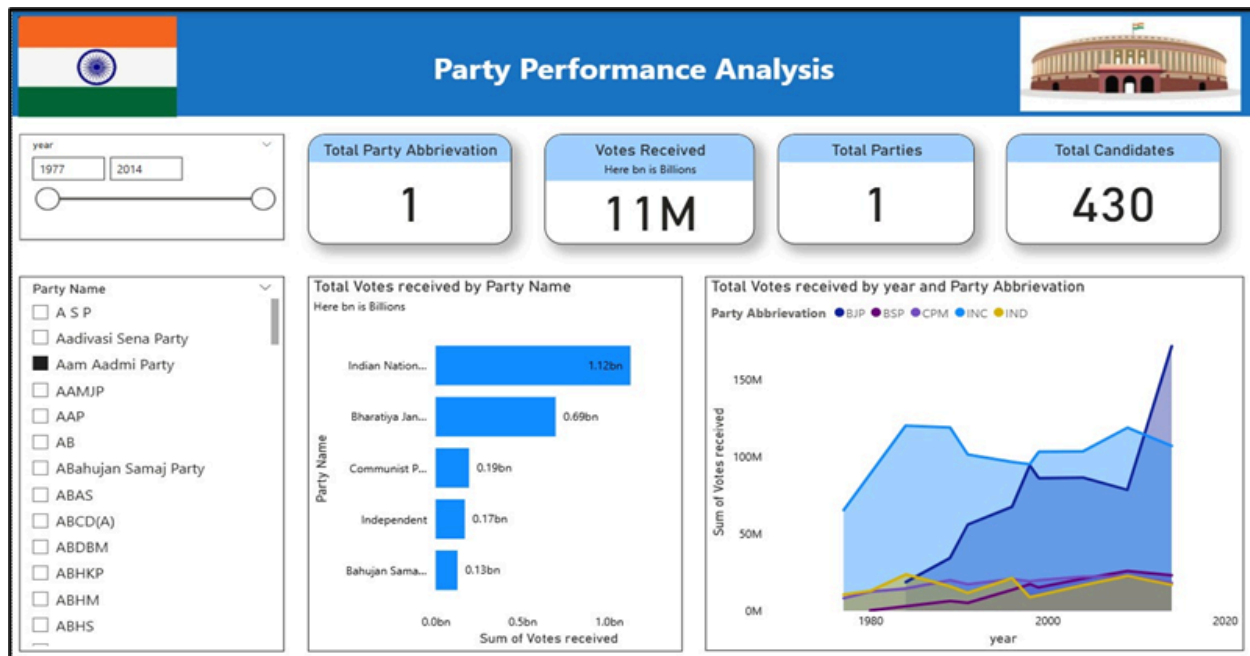
Male candidates dominate elections, indicating gender imbalance and scope for improved female representation.

Insight generated from this page:

- ❖ Votes received increases over year . So , an increasing trend indicates rising voter participation.
- ❖ Dense Populated area stands out in contribution of significantly more votes.
- ❖ Male candidates dominate elections (94.96%) , indicating gender imbalance and scope for improved female representation.

- ❖ Most parliamentary constituencies are General (84.39%), followed by SC (11.24%) and ST (4.37%).

6.2 Party Performance Analysis



Page Objective

This page focuses on how political parties performed, how votes were distributed among them, and how party strength evolved over time.

Data used : Part name , year , votes received

Visuals used : KPI Cards (to get quick summary) , year and party slicer to filter dashboard ,Area Chart (to get performance over year) ,bar graph (comparison).

1. Card: Count of Party Abbreviations (1068)

What it shows:

Total number of unique party abbreviations.

2. Card: Sum of Votes Received (4bn)

What it shows:

Total votes polled across all years.

Insight:

Demonstrates the massive scale of voter participation in Indian elections.

3. Card: Count of Party Names (1404)

What it shows:

Number of political parties that contested elections.

4. Card: Count of Candidates (56,601)

What it shows:

Total candidates who contested elections.

5. Bar Chart: Total Votes Received by Party Name

What it shows:

Comparison of votes received by major parties.

Insight:

National parties like INC and BJP dominate vote share, but regional parties still play a significant role.

6. Line Chart: Total Votes by Year and Party

What it shows:

Trend of votes received by major parties over time.

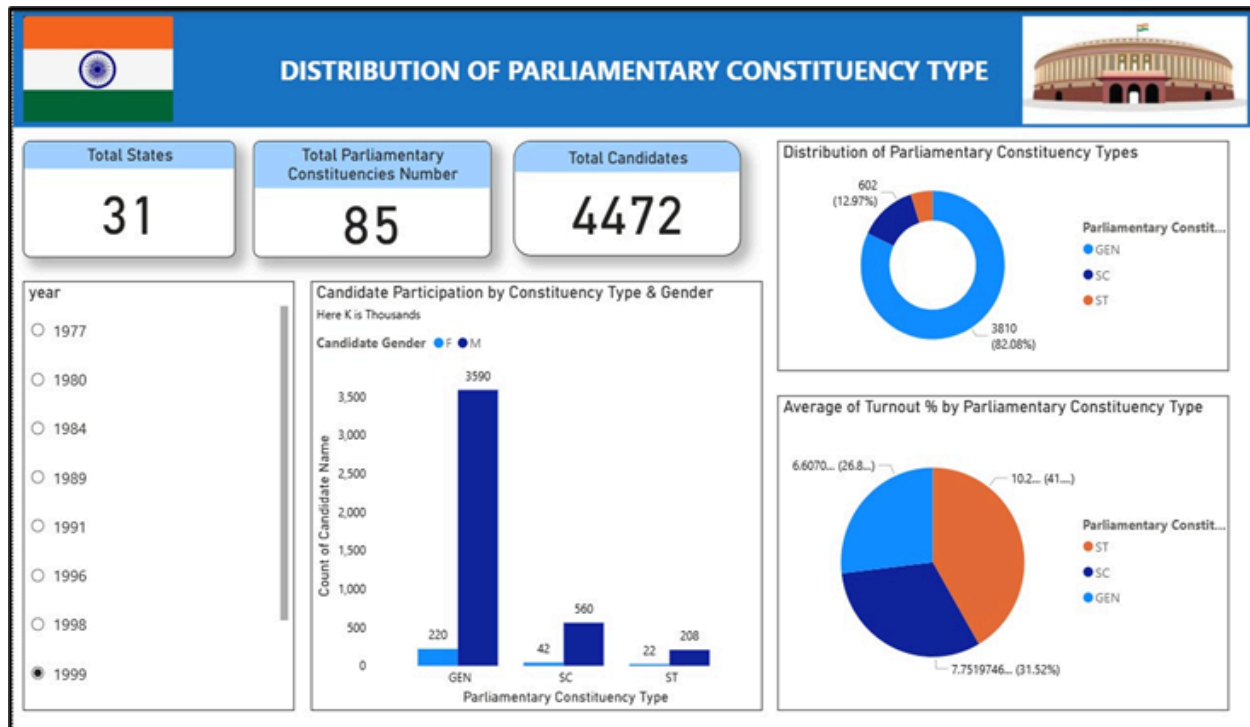
Insight:

Reveals shifts in political dominance, emergence of new parties, and decline of others.

Insight generated from this page:

- A large number of votes are received each election year but votes are concentrated among a small number of parties.
- National parties like INC and BJP dominate vote share, but regional parties still play a significant role.
- Indian National Congress received the highest votes from 1980 to 2009 , then in 2014 Bhartiya Janta Party received the highest vote.
- Party performance varies significantly across different years.

6.3 Distribution of Parliamentary Constituency Type



Page Objective

This page analyzes how constituency types differ in terms of count, candidate participation, gender representation, and voter turnout.

Data used : Parliamentary Constituency type , year , candidate gender , turnout %

Visuals used : KPI Cards (to get quick summary) , year slicer to filter dashboard ,bar graph (comparison of female and male winner candidate) , Pie Chart and Donut Chart (Distribution of number of constituency and turnout%).

1. Card: Count of State Name (35)

What it shows:

Number of states included for constituency-type analysis.

2. Card: Count of Parliamentary Constituencies (85)

(Filtered view based on selection)

What it shows:

Total constituencies under current slicer selection.

3. Donut Chart: Distribution of Parliamentary Constituency Types

What it shows:

Proportion of GEN, SC, and ST constituencies.

Insight:

Confirms that reservation policies are consistently applied across election years.

4. Clustered Column Chart: Candidate Participation by Constituency Type & Gender

What it shows:

Number of male, female, and other candidates contesting in each constituency type.

Insight:

Male dominance exists across all constituency types, while female participation remains limited, even in reserved seats.

5. Pie Chart: Average Turnout % by Constituency Type

What it shows:

Average voter turnout in GEN, SC, and ST constituencies.

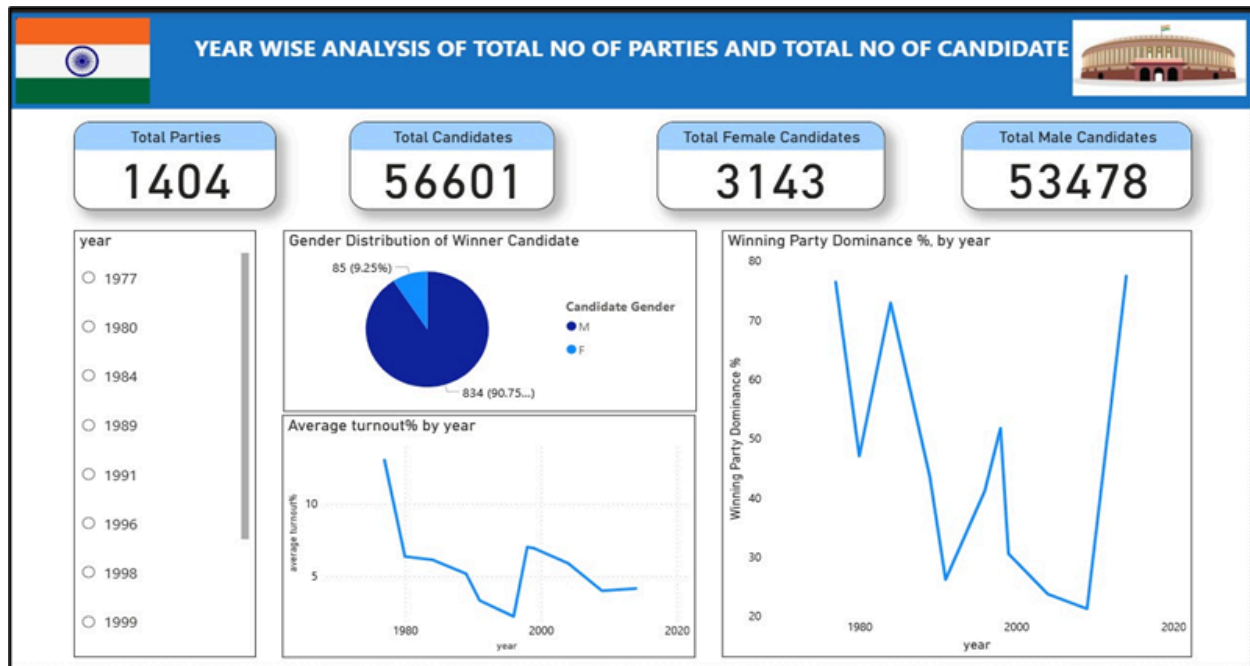
Insight:

ST constituencies often show higher turnout, indicating strong voter engagement in reserved regions.

Insight generated from this page:

- Male as well as female participation are kept on increasing each year but still female participation show slow progress.
- Male dominance exists across all constituency types, while female participation remains limited, even in reserved seats highlighting the need for women's participation.
- General constituencies dominate in number , SC and ST constituencies play a crucial role in representation and demonstrate voter participation.

6.4 Year Wise Analysis



Page Objective

This page analyzes year wise analysis Total number of parties and Total number of candidates.

Data used : Year , candidate gender , voter turnout % , constituency type along with DAX measure total no. of candidate

Visuals used : KPI Cards (to get quick summary of total candidates) , year slicer to filter dashboard ,line chart (trend over years of turnout % and winner dominance %) , Pie Chart (to show Gender Distribution in winning candidates).

1. Card: Count of Total number of parties

What it shows:

Represent Total no of parties present in the dataset and varies year by year

2. Card: Count of Total number of candidate

What it shows:

Represent Total no of candidates present in the dataset and varies year by year

3. Card: Count of Total number of female candidate

What it shows:

Represent Total no of female candidates present in the dataset out of total candidate and varies year by year

4. Card: Count of Total number of male candidate

What it shows:

Represent Total no of male candidates present in the dataset out of total candidate and varies year by year

5. Gender Distribution of Winner Candidate (Pie Chart)

What it shows:

Displays the proportion of male and female winning candidates.

Insight:

Highlights gender imbalance in election outcomes.

6. Average Turnout Percentage by Year (Line Chart)

What it shows:

This chart shows average Turnout % in each year

Shows trends in voter participation across election years.

Insight:

There was the highest turnout % in 1977 then it kept on decreasing and lowest in 1996 then it increased from 1998 and then became stable in 2014.

Helps analyze changes in public engagement over time.

7. Winning Party Dominance% (Line Chart):

What it shows:

This chart shows the Winning Party Dominance Index, which measures the percentage of constituencies won by the top party in each election year.

The tooltip/card highlights which party held that dominance, allowing us to clearly distinguish single-party dominance from coalition eras.

Insight:

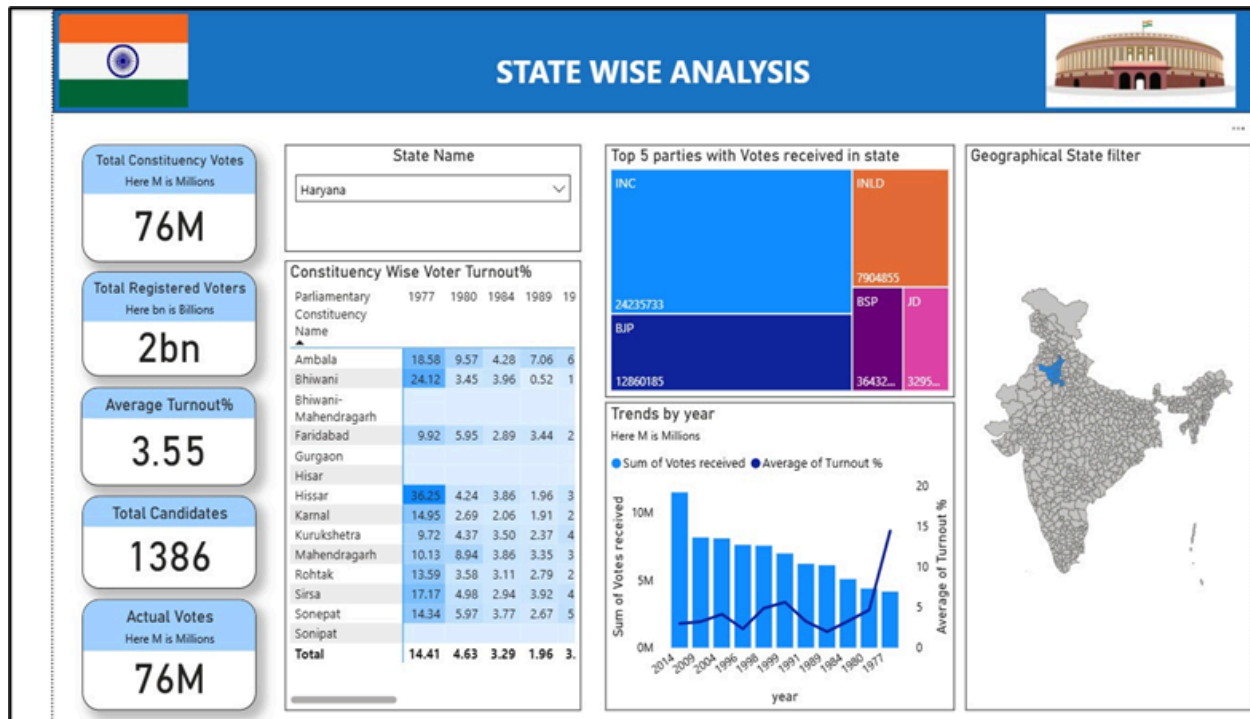
Higher values indicate a dominant single-party system, while lower values reflect coalition-based politics .

When observed alongside the number of participating parties, it clearly highlights the transition from dominance to coalition eras.

Insight generated from this page:

- There are a total 1404 parties with 56601 total candidates , 3143 female and 53478 male candidates out of total candidates.
- There was the highest turnout % in 1977 then it kept on decreasing and lowest in 1996 then it increased from 1998 and then became stable in 2014. It tells changes in public engagement over time.
- There is male dominance in winning candidates compared to female winning candidates.
- Winning Party Dominance% higher value tells about single-party dominance and lower value indicates coalition era. For example, the winning dominance % of BJP is 77% in 2014.

6.5 State Level Analysis



Page Objective

This page analyzes the State wise performance of the party.

Data used : State name , turnout% , votes received , year along with DAX measure actual and eligible voters.

Visual used : KPI Cards (to get quick summary total constituency , actual voters , eligible voters), state slicer to filter dashboard , Line Chart and bar chart (to find correlation between turnout % and votes received) , tree map (Votes received by each party) , table to get exact data of turnout % of each constituency) , map to filter the entire dashboard by selecting a specific state.

1. Card: Total Constituencies Votes

What it shows:

Represent Total Constituencies Votes

2. Card: Total registered voters

What it shows:

Represent Total no of registered voters as well as present in specific state.

3. Card: Average of turnout %

What it shows:

Represent Total average of turnout% as well as average turnout% of specific state.

4. Card: Total Candidates

What it shows:

Represent Total no of candidates as well as total candidates present in specific state.

5. Card: Actual Votes

What it shows:

Represent Total no of Actual Votes as well as present in specific state.

6. Shape Map – Geographical State Selection

What it shows:

An interactive map of India allowing users to filter the entire dashboard by selecting a specific state (e.g., Andhra Pradesh). Helps users visualize the geographical context of the data being analyzed. Selected regions are highlighted to show the physical distribution of constituencies

7. Treemap – Sum of Votes Received by Party

What it shows:

Illustrates the distribution of the total votes received across different political parties.

The size of each rectangle represents the party's vote share, making it easy to identify dominant parties like INC and TDP.

Useful for comparing the relative strength of multiple parties simultaneously.

8. Combination Chart (Line and Stacked Column) – Trends by Year

What it shows:

Displays the Sum of votes received as bars and the Average of voters_turnout_% as a line over different election years.

Ideal for identifying correlations between the volume of votes and the percentage of voter participation over time.

Dual axes allow for the comparison of two different scales (millions of votes vs. percentage) in one visual.

Insight:

This graph compares total votes received with the average voter turnout percentage across election years. It shows that increases in total votes do not consistently align with higher turnout percentages, indicating no clear positive relationship between voter participation and total votes.

9. Matrix Table – Constituency Wise Voter Turnout

What it shows:

Provides a detailed breakdown of voter turnout percentages for each constituency (e.g., Adilabad, Amalapuram) across various years.

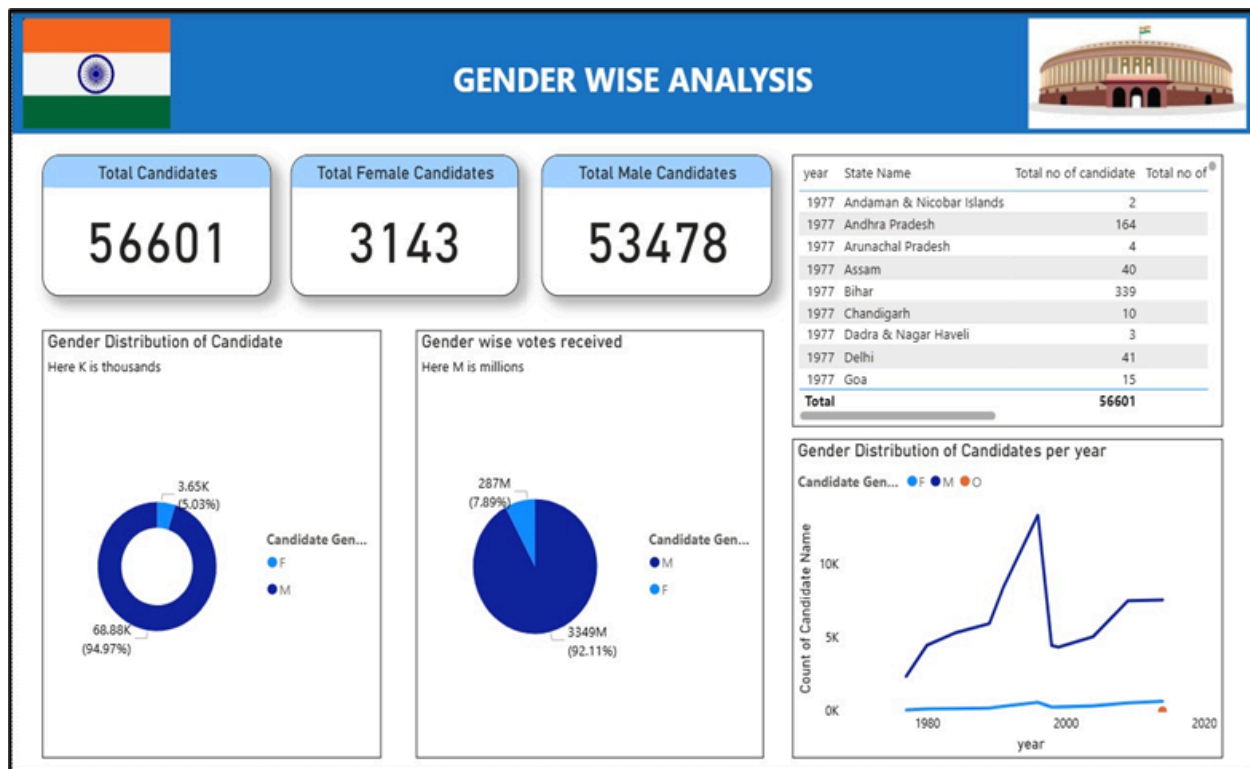
Allows for a deep dive into localized performance trends and historical comparisons.

Insight generated from this page:

- It shows that increases in total votes do not consistently align with higher turnout percentages, indicating no clear positive relationship between voter participation and total votes.

- The Treemap reveals that the vast majority of the total votes received concentrated towards two or three major parties .

6.6 Gender Wise Analysis



Page Objective

This page analyzes gender wise analysis .

Data used : Candidate gender , votes received , year along with DAX measure total candidates .

Visuals used : KPI Cards (to get quick summary of total candidates), Line Chart (To show changes in participation over years) , table to get exact data of total no of candidates in each year) donut chart and pie chart (to show Gender distribution in votes received and number).

1. Card: Count of Total number of candidate

What it shows:

Represent Total no of candidates present in the dataset and varies year by year.

Total no of candidate = `DISTINCTCOUNT('indian-national-level-election'[Candidate Name])`

2. Card: Count of Total number of female candidate

What it shows:

Represent Total no of female candidates present in the dataset out of total candidate and varies year by year.

Total no of female candidate =
`CALCULATE(DISTINCTCOUNT('indian-national-level-election'[Candidate Name]),'indian-national-level-election'[Candidate Gender]="F")`

3. Card: Count of Total number of male candidate

What it shows:

Represent Total no of male candidates present in the dataset out of total candidate and varies year by year.

Total no of male candidate =
`CALCULATE(DISTINCTCOUNT('indian-national-level-election'[Candidate Name]),'indian-national-level-election'[Candidate Gender]="M")`

4. Gender-wise Votes Polled (Pie Chart)

Why used:

To compare **how votes are distributed** between male and female candidates.

What it presents:

- Male candidates receive the majority of votes

- Female candidates receive a much smaller share

This shows that lower participation of women also reflects in lower vote share.

5. Donut Chart: Gender Distribution of Candidates

What it shows:

Proportion of male, female, and other candidates.

Insight:

Male candidates dominate elections, indicating gender imbalance and scope for improved female representation.

6. State-wise Table

What it shows:

Tables are best when exact numbers matter.

What it presents:

- State-wise count of total candidates
- Number of male and female candidates in each state

This helps identify which states have relatively better or worse female representation.

7. Male vs Female Candidates Over Time (Line Chart)

What it shows:

Line charts clearly show trends over time.

What it presents:

- Male candidate participation has increased over the years
 - Female participation has improved slightly but remains very low
-

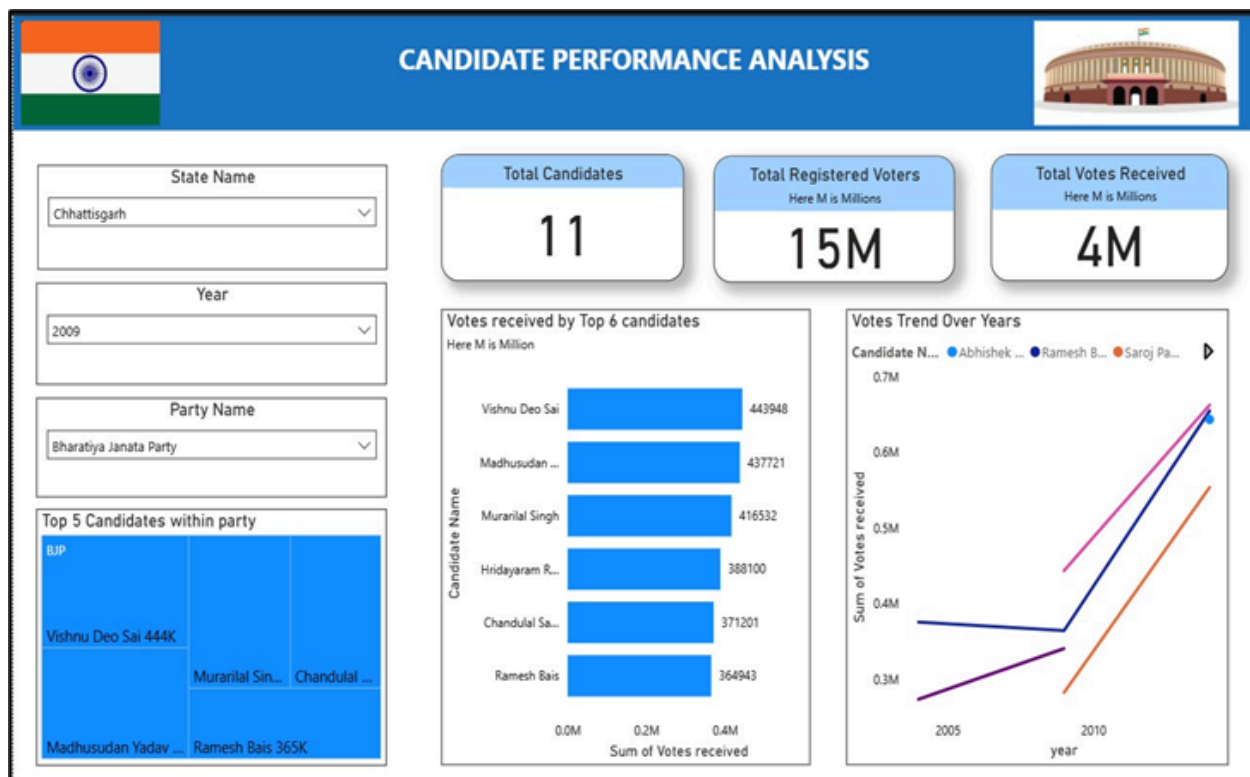
Conclusion:

This page clearly shows that Indian elections are heavily male-dominated. While there is slow progress in female participation, gender imbalance remains a major challenge in electoral representation.

Insight generated from this page:

- Male dominant in election both in numbers and total votes received.
- Male participation also increased faster than female participation over years .
- Although female representation shows minor improvement over time, but still low.
- The Gender Wise Analysis highlights the need for stronger efforts to improve women's participation in Indian politics.

6.7 Candidate Performance analysis



Page Objective

This page analyzes Candidate performance analysis.

Data Used: State , year , candidate name and gender , votes received.

Visuals used : KPI Cards (to get quick summary of total candidates , total votes received and total voters), State , Year and party name slicer to avoid missing of data and to focus on particular context), bar chart (to show top candidate) , Tree map to show top candidate received more votes within party and line chart (to display candidate performance over years).

1. Slicer: State

What it shows:

Allows to slice the entire dashboard according to any state

2. Slicer: Year

What it shows:

Allows to slice the entire dashboard according to any year

3. Slicer: Party Name

What it shows:

Allows to slice the entire dashboard according to any party.

4. Card: Total Votes Received

What it shows:

Represent Total Votes Received by party in specific year or state.

5. Card: Total Candidates

What it shows:

Represent Total number of candidates belong to that party in specific year or state

6. Card: Total Registered Voters

What it shows:

Represent Total Votes Registered voters in a specific year or state.

6. Stacked Bar Chart : Votes received by Top 6 candidates

What it shows:

This chart shows that sum of votes received by Top 6 candidates

Insight:

Compares individual performance side-by-side using the sum of votes received.

7. Treemap – Sum of Votes Received by Party

What it shows:

This visual shows how votes are distributed among top candidates of the same party.

Larger blocks indicate candidates with higher vote share, making it easy to identify star performers.

8. Line Chart – Votes Trend Over Years

What it shows:

This line chart shows how votes for key candidates have changed over time.

It helps us understand consistency, growth, or decline in a candidate's popularity.

Insight generated from this page:

- A few candidates receive a large share of votes, while others contribute less, showing uneven performance within the party.
- Certain candidates remain strong across years, indicating stable voter trust.
- Even with many registered voters, votes are concentrated among top candidates.

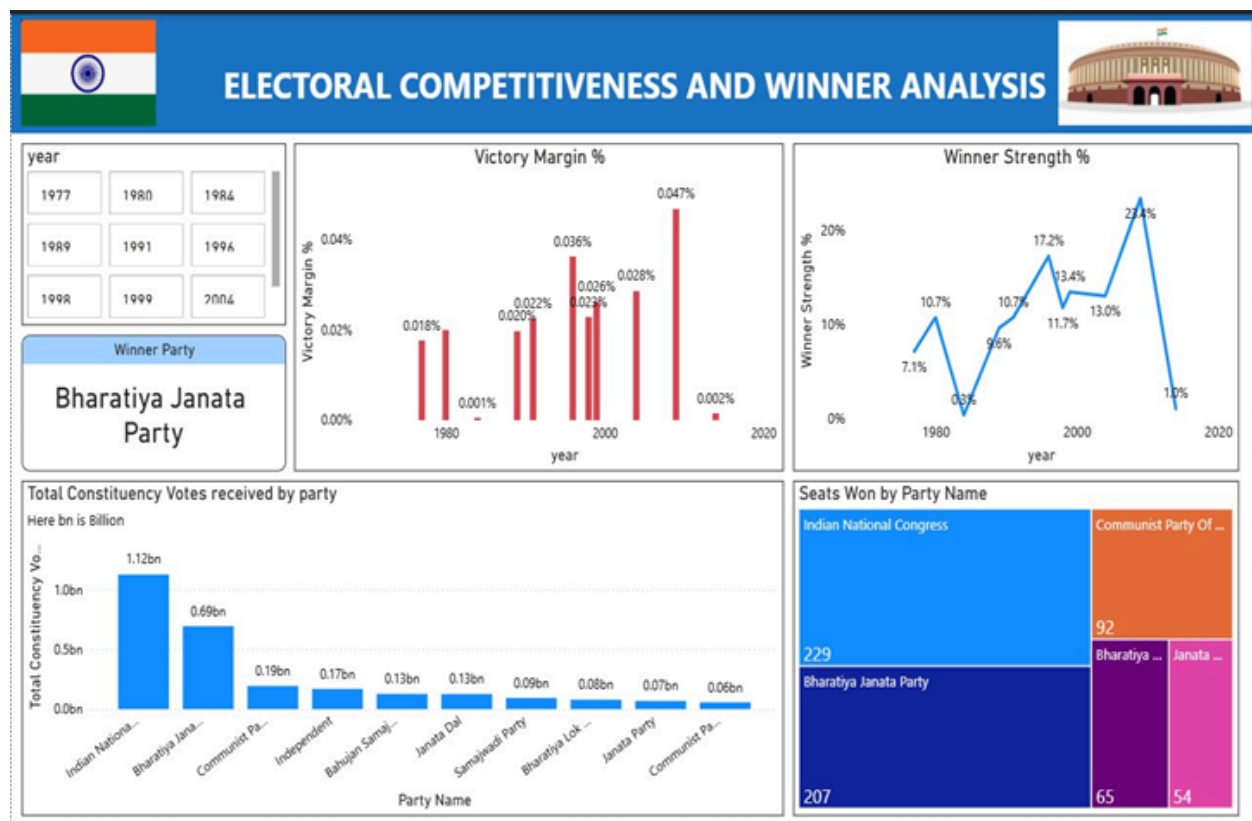
Conclusion:

Overall, this page shows that **candidate performance is not equal within a party**.

A small number of strong candidates drive most of the votes, while others play a supporting role.

This highlights the importance of **strong individual leaders in elections**, not just party strength.

6.8 Electoral competitiveness analysis



Page Objective

Electoral competitiveness analysis uses victory margins to differentiate closely fought elections from landslide wins and reveals regional patterns of political dominance and voter behavior.

Data Used: Part name , state ,year , votes received along with DAX measures winner strength , victory margin % , total constituency votes , seats won.

Visuals used : KPI Cards (to display winner party and seats won by winner party), year slicer (to filter dashboard) , Line Chart (To display trend of winner strength % over years) , bar graph (to display victory margin % in each year and total constituency votes by each party) and tree map (Top 5 parties with highest seats won).

1. Year slicer

What it shows:

It helps to filter the entire dashboard year wise.

2. Victory Margin % (Column Chart)

What it shows:

Represent How big was the win compared to all voters who participated?

Small % → highly competitive election

Large % → one-sided contest

$$\text{Victory Margin \%} = \text{DIVIDE}([\text{Winner Votes}] - [\text{RunnerUp Votes}], [\text{indian-national-level-election'}[\text{Total Valid Votes}]])$$

$$\text{Total Valid Votes} = \text{CALCULATE}(\text{SUM}([\text{indian-national-level-election'}[\text{Votes received}]], \text{ALLEXCEPT}([\text{indian-national-level-election'}, \text{indian-national-level-election'}[\text{year}], \text{indian-national-level-election'}[\text{Parliamentary Constituency Number}]])$$

Insight:

The victory margin percentage remains consistently low across most election years, indicating that Indian parliamentary elections are highly competitive. Several years show extremely small margins, highlighting closely contested constituencies where minor vote swings could change the outcome.

3. Winner Strength % (Line Chart)

What it shows:

Represent By what fraction did the winner outperform the runner-up?

Small % → narrow win

Large % → dominant win

Winner Strength % = $\text{DIVIDE}([\text{Winner Votes}] - [\text{RunnerUp Votes}], [\text{Winner Votes}])$

Insight:

Winner strength fluctuates significantly across election years. Some years reflect dominant victories with high winner strength, while others show weak dominance, indicating fragmented voter preferences and strong opposition presence.

4. Total Constituency Votes received by party

What it shows:

Cumulative votes received by each political party across all constituencies and years.

Total Constituency Votes Received by party =
 $\text{CALCULATE}(\text{SUM}(\text{'indian-national-level-election' [Votes received]}),$
 $\text{ALL}(\text{'indian-national-level-election' [Candidate Name]},$
 $\text{'indian-national-level-election' [Party Abbrievation]}))$

Insight:

National parties such as the Indian National Congress and Bharatiya Janata Party have received significantly higher total vote shares compared to regional parties, reflecting their broader national presence and voter base. However, regional parties still contribute substantial vote volumes in specific regions.

5. Seats Won by Party

What it shows:

Total number of parliamentary seats won by each party. It shows rise and fall of political parties.

Seats Won = $\text{SUM}(\text{'indian-national-level-election' [Winner]})$

Insight

Seat distribution is uneven across parties. While a few major parties dominate in terms of seats won, smaller and regional parties maintain electoral relevance by securing representation in specific constituencies, indicating a multi-party democratic structure.

Insight generated from this page :

- Winner strength % fluctuates, meaning dominance is not consistent across years.
- Victory Margin % fluctuates, There is high competition in some years and in some years there is low competition.
- Some parties win more seats even with small vote shares, and some parties win low seats with high votes share showing high seats does not mean high votes received.

Conclusion:

Under the First Past the Post system, winning a seat depends on securing the highest votes in each constituency, not on overall vote share.

As a result, a party can win more seats even with fewer total votes, while another party may receive a large number of votes but win fewer seats.

For example BSP in 2014 received 23M votes but won 0 seats.

7.Challenges Faced:

Challenges and their resolution during the 4 milestone are:

7.1 Data Integration and Cleaning:

Consolidating raw election records required handling scientific notation in turnout percentages and ensuring party names were standardized across different years. This was essential to prevent duplication in charts like the "Sum of votes received by partyname".

7.2 Technical Alignment in Power BI:

Navigating the updated Power BI interface to locate specific field buckets like **Value**, **Legend**, and the **Filters** pane proved challenging for a beginner. Correcting the logic to show only winners (where **Winner = 1**) was a critical step for KPI accuracy.

7.3 Communication and Time Management:

Organizing the project into a strict 5-week timeline required balancing dataset understanding with the technical development of complex visuals. Ensuring that the documentation and analysis were finalized by Week 5 was a key priority.

7.4 Dashboard Optimization and Performance Issues:

Managing a dataset with over 50,000 candidate records required optimizing visuals like the "State-wise Analysis" table to ensure smooth filtering. Ensuring that the Treemaps and Line charts updated instantly across different election years was a technical focus.

7.5 Design Consistency and Storytelling Alignment:

Aligning the visual elements, such as blue headers, borders, and shadows, was necessary to create a professional and "exact" look across both pages. The main goal was to ensure the gender analysis and winner metrics told a clear, unified story of India's electoral history.

8. Learnings & Skills Acquired :

8.1 Technical Skills:

Gained hands-on experience with **Power BI Desktop**, including effective use of the Visualizations, Data, and Filters panes to create interactive dashboards.

- Power Query : Gained knowledge about loading dataset , cleaning errors and converting data into usable formats.
- DAX (Data Analysis Expression) : Mastering the formula language to create custom calculations ,measures etc.
- Dashboard & Report Design : Gained skills in selecting right visuals and implementing interactive slicers.

8.2 Analytical and Problem-Solving Skills:

- Developed logical filtering and calculation techniques to extract key KPIs, such as identifying winners using the *Winner = 1* condition and calculating runner-up votes for accurate victory margin analysis.
- Applied comparative analysis to understand trends like party dominance, gender participation, and vote distribution across years.
- Transformed large volumes of raw election data into meaningful insights by designing measures such as Victory Margin %, Winner Strength %, and Dominance Index.
- Strengthened problem-solving ability by resolving issues related to incorrect aggregations, null values, percentage formatting, and context handling in DAX.
- Trend and pattern recognition: Identifying anomalies , correlations and hidden patterns.
- Business Problem Solving: Translating business questions into data driven insight
- Data Storytelling The ability to shape visual insights into narrative that stakeholder can easily understand.

8.3 Soft Skills and Team Collaboration:

Practiced disciplined time management by adhering to a structured 8-week roadmap, progressing from initial data cleaning to final insight validation.

- Enhanced technical writing skills by documenting the dashboard's architecture, challenges, and user instructions for a beginner-friendly presentation.
- Improved communication skills through structured presentation and explanation of analytical findings.

8.4 Domain Knowledge and Application

Acquired deep insights into the Indian electoral landscape, specifically tracking the historical participation and performance of male versus female candidates from 1977 to 2014.

- Understood the significance of victory margins and voter turnout percentages in determining the competitiveness of different election years.
- Learned how seat distribution and vote share reflect political dominance and competition.

9. Testimonials from Team:

1. Arya Mishra :

Collaborating on this project was a highly integrated experience where we functioned as a cohesive unit. I was deeply involved in the initial dataset research and the iterative cleaning process to ensure our Power BI visuals were accurate. Beyond the technical development of the 'Winner Analysis' page, I also contributed to the structured documentation to ensure our project's story was clearly communicated.

2. Kritika Balpande:

Our team maintained a hands-on approach across every phase of the project, from the first week of data collection to the final dashboard optimization. I focused on ensuring that our gender-wise analysis visuals were both informative and aesthetically consistent with our professional design standards. Working together on the final presentation helped us synthesize our technical findings into a compelling narrative about electoral trends.

3. Radha sri Yakkala:

This project succeeded because we shared the responsibility for every critical task, including data transformation and building complex Power BI measures. I took a lead role in verifying the integrity of our KPI cards while also collaborating closely on the PowerPoint slides and project documentation. This end-to-end involvement allowed us to tackle challenges like design consistency and data integration as a unified team.

10.Conclusion:

ElectViz successfully bridges the gap between raw electoral data and actionable public information by providing a centralized, automated visualization framework. By transforming complex datasets into intuitive, interactive dashboards featuring choropleth maps and swing-analysis charts, the platform overcomes the high latency and manual errors inherent in traditional reporting. The implementation of a robust data pipeline ensures that real-time insights enables newsrooms to identify critical "voter-shift" and "anti-incumbency" patterns instantly. Ultimately, this project offers a scalable, low-latency solution that enhances democratic transparency and provides broadcast-ready graphical insights for the digital age.

Through effective data preprocessing, calculated measures, and interactive dashboards, raw election data was transformed into clear and actionable visual stories. The dashboards highlight how votes translate into seats under the First-Past-The-Post system, how competitive elections are across years, how victory margins differ, and how candidate and gender representation has evolved over time.

The project also demonstrates the importance of choosing correct metrics and visuals to avoid misleading interpretations, especially when dealing with large-scale public data. By using slicers and filters, users can explore insights at state, party, year, and candidate levels, making the analysis flexible and user-friendly.

Overall, this project shows how data analytics and visualization tools like Power BI can support transparency, better understanding of electoral outcomes, and informed discussions about democracy. It successfully bridges the gap between raw election data and meaningful insights, making complex election trends easy to understand for both technical and non-technical audiences.

11.Acknowledgements:

Institutional Appreciation: Infosys Springboard

We extend our sincere gratitude to **Infosys Springboard** for providing a robust and comprehensive learning ecosystem. The platform's wealth of resources and structured curriculum served as the cornerstone of our professional development, offering us the tools and environment necessary to translate theoretical knowledge into a tangible, high-impact project.

Technical Mentorship

We are thankful to our **Mentor Mrs Nityasri S J** for their invaluable guidance throughout the project's lifecycle. Beyond mere supervision, your technical oversight was crucial during the more intricate phases of development. Your constructive feedback significantly shortened our learning curve within the **Power BI interface**, particularly in mastering complex data modeling and refining our visualizations to ensure they were both insightful and user-centric.

Team Collaboration

This project is a true reflection of the **collective dedication of our team**. The success of this endeavor was built on hours of collaborative effort, specifically in the rigorous stages of **data cleaning and preprocessing**. Each team member's contribution to searching for reliable data sources, troubleshooting technical hurdles, and synthesizing our findings into final documentation and presentations was indispensable.

The Power BI Community & Technical Resources

Finally, we would like to acknowledge the broader **Power BI community** and the wealth of online technical documentation. These resources were vital in helping us overcome specific challenges related to **dashboard optimization**, ensuring design consistency, and implementing best practices for data storytelling.